Lecture 14: What Motivates Us: Sex

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-14 (CC BY-NC-SA 3.0).

Before you start

This lecture looks at sex from an evolutionary and a scientific point of view. Professor Bloom explains why males and females of many species differ, what research shows about human sex differences and attractiveness, and what is known about sexual orientation.

Learning goals

After this lecture you should be able to:

- Explain why evolutionary facts do not settle moral questions, and why genetic causes are not always fixed.
- Explain Trivers' idea of parental investment and use it to predict size differences between males and females.
- Describe evidence about human sex differences in aggression, choosiness and mate preference.
- Describe what makes a face attractive, including the "average face" finding.
- Describe social factors behind some sex differences, and what is known about sexual orientation.

Key ideas

1. Addressing Morality and Inevitability on Evolution, Sex and Gender

- James Gleick reports that Americans say sex is their favorite activity, but the average time per day devoted to it is four minutes and three seconds. Still, much of life follows from it: marriage, family, children, competition, art and music.
- Morality: Steve Pinker writes, "Nature does not dictate what we should accept or how we should live our lives." Bloom adds that many "unnatural" things, such as flying in a plane, are not wrong. Nothing moral necessarily follows from biology.
- Inevitability: Richard Dawkins asks why genes are seen as more fixed than "television, nuns or books." Bloom's poor eyesight is genetic but easy to fix with glasses. Attitudes towards obese people are cultural but very hard to change. So genetic does not mean inevitable, and cultural does not mean easy to fix.

2. Basic Sex Ed

- Biologists define males as animals with a little sex cell (sperm, which carries only genes) and females as animals with a big sex cell (which also has food and a protective cover).
- Puzzle: why are males of many species bigger and more aggressive? Robert Trivers' answer is "parental investment": any investment that increases an offspring's chance of survival at the cost of

the parent's ability to invest in other offspring.

- Females usually invest more. While pregnant with one offspring, a female cannot have another. So males compete with other males (larger size, special weapons). Females are choosy, so males also evolve displays, like the peacock's feathers.
- Male pipefish carry the eggs in a pouch, and the females are larger and compete more. Emperor penguin parents invest equally and are about the same size. Male elephant seals compete for harems and are four times bigger than females. Gibbons are fairly monogamous, raise children together, and are about the same size.

3. Sex Differences among Humans

- Most human cultures are polygamous; American culture has "serial monogamy." The average man is about fifteen percent larger than the average woman, which suggests some male-male competition in our evolutionary history.
- Aggression: on average, males are more physically violent in the womb (they kick more), as children, and as adults. Prisons are mostly full of men. Testosterone increases aggressiveness. These are averages: there are violent women and non-violent men.
- Choosiness: the theory predicts that males will be more open to sex with strangers. Prostitution is a near-universal male interest, and pornography is a human universal.
- In surveys about how many sexual partners they want, women answered less than one in the next month, one in two years, and four to five in a lifetime. Men answered two, eight and eighteen.
- In a study done about ten years earlier, attractive actors asked strangers on a campus three questions. Over half of the women agreed to a date, very few agreed to visit the apartment, and none agreed to sex. Men agreed 50%, 69% and 75% of the time.
- A study of 10,000 people in thirty-seven countries found that everyone says they value kindness and intelligence. Women focus more on power, status and interest in investing in children. Men focus more on the ability to have children.

4. Beauty: An Average Face

- Beauty seems to signal youth (large eyes, full lips, smooth skin) and health (no deformities, clear eyes, unblemished skin, intact teeth, and an average face).
- An average face is attractive because deformities are variations from the average. When many faces are morphed together by computer, the result looks better and better.
- Babies also prefer to look at average faces. This suggests the preference is to some extent hard-wired, not only a product of culture.
- Faces can look better than average. One example is a "feminized" average female face, with features made more feminine. Shown two male faces, most people prefer face B, but women who are ovulating prefer face A, which has a chiseled jaw and square chin. This has been replicated in three labs; the explanation is complicated.

5. Social Factors for Sex Differences

- Some sex differences, such as in aggression and mate preference, seem universal and are likely biological adaptations. For others, Bloom is "personally quite skeptical" that they reflect biology.

- People treat a baby differently if they think it is a boy or a girl. An application or paper is judged differently with the name "John Smith" or "Joan Smith."
- From about age four to eleven, boys mostly stay with boys and girls with girls. Eleanor Maccoby proposed that this self-segregation may exaggerate small differences.
- Simon Baron-Cohen, in The Essential Difference, argues that women are by nature more empathetic. Problems of social cognition, such as autism, are much more frequent in males.
- Larry Summers, then president of Harvard, suggested that "intrinsic aptitude" and greater male variability help explain why fewer women are in science. This was extremely controversial. Bloom points students to a debate between Steve Pinker and Liz Spelke.

6. Sexual Orientation

- Roughly 98% of women are attracted to men and 96% of men to women, but these numbers are hard to estimate. Orientation is about what you want, not only behavior. The real puzzle is exclusive homosexuality.
- It is almost certainly not a choice: few people decide whom they are attracted to. It also does not seem to come from experiences after puberty; differences may appear before puberty.
- Twin studies show some genetic predisposition. But if one identical twin is gay, the chance that the other is gay is about fifty percent, not one hundred. So some experience, possibly prenatal (before birth), must also matter.
- From an evolutionary view, exclusive homosexuality is a mystery, because genes that lead to having no offspring should be weeded out. Bloom stresses that this carries no moral weight.

7. Question and Answer on Sex and Gender

Many animals engage in same-sex sexual behavior, but Bloom did not know whether exclusive homosexuality exists in other species.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
inevitable	cannot be avoided or changed	Bloom's eyesight is genetic, but glasses fix it
parental investment	effort that helps one offspring survive but limits having others	A female cannot have another offspring while pregnant
polygamous	having more than one mate	Most human cultures are polygamous
serial monogamy	one partner at a time, one after another	American culture
testosterone	a male sex hormone	It increases aggressiveness
choosy	careful when choosing a mate	Females are biologically choosy
average face	a face made by combining many faces	Babies prefer to look at average faces
gender self-segregation	children separating into groups of their own sex	Boys with boys, girls with girls, from about age four to eleven

Term	Meaning in simple English	Example from the lecture
empathy	understanding the feelings of others	Simon Baron-Cohen's The Essential Difference
sexual orientation	whom a person is sexually attracted to	About 96% of men are attracted to women
prenatal	before birth	Prenatal experience may affect orientation

Check your understanding

1. What is Pinker's point about nature and morality?
2. How does Bloom's eyesight example show that "genetic" does not mean "inevitable"?
3. What is parental investment, and why is it usually higher for females?
4. Male pipefish carry the eggs. What does parental investment theory predict about pipefish females?
5. Why are male elephant seals much bigger than females, while gibbons are about the same size?
6. What evidence did Bloom give that, on average, men are less choosy than women?
7. Why might an average face look attractive? What do babies' preferences suggest?
8. What does the fifty percent figure for identical twins tell us about sexual orientation?

Answer guide

1. Nature does not tell us how to live. Something can be a product of evolution without being right, and "unnatural" things are not necessarily wrong.
2. His poor eyesight is caused by his genes, but it is easy to fix with glasses or contact lenses.
3. It is any investment that helps an offspring survive but reduces the parent's ability to invest in other offspring. While pregnant, a female cannot have another offspring.
4. The females should be larger and compete more, and this is true.
5. Male elephant seals compete for harems ("winner take all"), so big males win. Gibbons are fairly monogamous and raise children together.
6. Survey answers about how many partners people want, the campus study with actors, and the universality of prostitution and pornography as male interests.
7. Deformities are variations from the average, so an average face shows that nothing bad has happened to it. Babies prefer average faces, so the preference is partly hard-wired.
8. It is high compared with the population, so genes play a role. But it is not 100%, so sexual orientation is not entirely genetic.

English practice

Useful phrases from the lecture

- "It turns out there is data on this." - introduces evidence for a question.
- "The point here is..." - gives the main message of an example.
- "This raises the question then, ..." - leads to the next question.
- "I am personally quite skeptical about..." - politely says you doubt a claim.

Speaking task (with your AI tutor)

- Explain parental investment theory in 4-6 sentences. Use: *parental investment*, *competition*, *choosy*.

Writing task

- Write one paragraph (80-120 words) explaining Bloom's idea that genetic does not mean inevitable, and cultural does not mean easy to fix. Use his two examples.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 14. Ask me questions about parental investment, one at a time."
- "Quiz me on the key vocabulary of Lecture 14."
- "I will explain why an average face is attractive in my own words. Check my understanding and my English."

Readings from the original course

- Gray, Peter. Psychology (5th edition), pp. 76-82, 195-201
- Angier, Natalie. "Woman: An Intimate Biography." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 255-262

My notes